

Designing a Dynamic Nation Simulation Event System

What the source games actually do

The strongest lesson from **NationStates** is not just that it has lots of stats. It is that the player shapes the country through broad, recurring policy choices rather than direct number-tweaking, and those choices feed a very wide statistical ecosystem. The official FAQ says issues appear several times a day, “how you respond determines almost everything about how your nation evolves,” and the game deliberately avoids micro-management. At the same time, the official World Census exposes an unusually broad spread of measures, including Civil Rights, Economy, Political Freedom, Population, Corruption, Crime, Health, Human Development Index, Public Education, Public Healthcare, Scientific Advancement, Tourism, Welfare, Wealth Gaps, and many industry-specific sectors. The official API even supports querying scores, ranks, and historical values for census scales, which is a useful reminder that trend lines matter almost as much as current values in a nation sim. ¹

NationStates is also valuable because it is **state-dependent rather than fully deterministic**. Its FAQ explicitly notes that the same issue can have different visible effects in different nations depending on the nation’s current condition, and NationStates staff have explained that visible stats sit on top of additional hidden stats and sub-traits behind the scenes. In practice, that means the same answer can reinforce the status quo in one country, cause a large ideological swing in another, or even change the same visible stat in different directions depending on the nation’s broader configuration. That is exactly the kind of “alive” feeling you want if your game should react to player history rather than treat every decision as a flat modifier. ²

NationStates also shows how to handle **eligibility and unlocks**. A moderator described issue assignment as “mostly random” from the set a nation qualifies for, with specific overrides for certain unlock issues; the same thread notes that every issue answered can change what the nation is eligible to receive next. The FAQ also shows that fields such as capital city, leader, religion, and even custom pretitle unlock only when population thresholds are reached, and community-maintained issue guides document “easter egg” issues gated by things like motto, pretitle, animal, season, or prior issue history. Even if you do not copy those exact rules, the pattern is excellent: keep a large content pool, filter by eligibility, then pick from the eligible subset with some randomness and some special-case priority. ³

The strongest lesson from **Democracy 2** is different. Its Steam page describes a simulation of “the motivations, loyalties and desires of everyone in the country,” built around policy sliders, voter blocs, civil-liberties tradeoffs, diplomacy, and budget pressure. Cliff Harris’s own Democracy 2 tips add two especially important ideas: many policy effects take time to filter through the simulation, and voters belong to multiple overlapping groups rather than a single clean class. In other words, Democracy’s power comes from *causal structure*: a policy does not merely add +5 to GDP; it ripples through group membership, issue salience, implementation lag, and public reaction. ⁴

That causal structure became more explicit in the series' later official modding documentation, and it is still directly relevant because the developer describes Democracy 1 through 4 as variations of the same homebrew neural-network-like simulation. In Democracy 4's official modding docs and blog posts, policies, events, voter groups, situations, and dilemma options are all treated as connected objects loaded from CSV or text files; equations can include inertia, cross-stat scaling, and decaying temporary effects. Events are evaluated by scored influences, dilemmas are chosen by likelihood and cooldown, and situations use start/stop triggers plus ongoing outputs while active. The official docs also expose pre-requisites that can block content unless conditions are met, such as coastline, land border, royalty, or tech prerequisites. ⁵

Academic work on interactive drama points toward the same synthesis you are asking for: a **hybrid architecture** that balances authored control with systemic emergence. A review of agency architectures frames the core design problem as balancing emergence and authorial control, while an event-centric planning paper argues for events with preconditions, postconditions, costs, and centralized coordination so that dynamic play can still conform to authored structure. That is a strong fit for a nation sim: not pure branching narrative, and not pure spreadsheet randomness, but authored event content sitting on top of a systemic simulation. ⁶

A stat model that borrows the best of both

The cleanest way to combine NationStates and Democracy is to use **three layers of simulation**.

The first layer should be a **small visible core dashboard**. The game needs a handful of numbers the player always watches because they summarize national health and political survival. A practical always-visible set is: **Population, Economic Output, Treasury or Debt, Employment, Stability, Liberty, Authority, Education, Technology, Health, Crime, and Diplomacy**. That mix reflects the macro focus of Democracy's economy-and-voter simulation, while preserving NationStates' broader nation-identity model where public services, freedom, social conditions, and international standing all matter. NationStates' census categories and Democracy's emphasis on GDP, unemployment, crime, liberties, and diplomacy all point in this direction: keep the watchlist small, but let the simulation beneath it be rich. ⁷

The second layer should be **secondary visible panels** that are not on the top bar but can be opened when the player drills down. This is where you put numbers that matter a lot but are not always worth screen real estate: corruption, violent crime, organized crime, inequality, inflation, food security, infrastructure, research throughput, administrative capacity, public trust, war exhaustion, migration pressure, black market activity, pollution, magical contamination, or orbital capability. NationStates shows the value of a wide statistical ecology, while Democracy shows that cross-linking many secondary values creates fertile ground for dilemmas and ongoing situations. ⁸

The third layer should be **hidden simulation variables**. This is where your game starts to feel less gamey and more alive. Instead of making Liberty a single scalar directly changed by every policy, define it as an aggregate of privacy, speech, due process, assembly, and religious freedom. Instead of making Crime one bar, compute it from petty crime, violent crime, organized crime, corruption, cybercrime, magical crime, or border raiding depending on era and setting. Technology can similarly be an aggregate of theory, engineering, industrialization, scientific institutions, and practical deployment. This mirrors NationStates' hidden sub-traits and Democracy's equation-driven interconnected objects, but it gives you more designer control and clearer tuning targets. ⁹

A useful rule is that **core stats should be aggregates, while secondary and hidden stats should be causes**. If Liberty drops, the player should be able to inspect why: perhaps emergency powers reduced due process, police militarization lowered privacy, and censorship harmed speech. If Crime rises, the real cause may be youth alienation, black-market profitability, war demobilization, or underfunded policing. This gives your events much better hooks, because they can target either the visible “headline” stat or the underlying cause.

You also asked for stats that are **not visible until the era or technology makes them meaningful**, and this is one of the best opportunities to improve on both inspiration games. Use two concepts here: a **latent value** and a **visible stat**. For example, your game can quietly accumulate a hidden “advanced research capacity” long before the player unlocks Rockets. Once Rockets are researched, that hidden capacity can be remapped into a visible **Space Program** stat. The same applies to **Magic**: arcane affinity, superstition, or anomalous incidents can exist before “magic” becomes a disclosed system; once the relevant discovery, religion, faction contact, or world rule is reached, the hidden precursor becomes a visible magical stat and starts unlocking magic-tagged events. NationStates already uses unlock thresholds and “awaiting legislation” states for some nation fields, while Democracy uses explicit prerequisites for conditional content. Your game should generalize that idea to all optional subsystems. ¹⁰

An event architecture for policies, diplomacy, war, and development

To keep the world feeling alive, treat “events” as **several related content types**, not one giant bucket.

A good structure is to separate them into **headlines, dilemmas, situations, and chains**. Headlines are passive developments that the player learns about but does not directly choose, very similar to Democracy events. Dilemmas are explicit choices with two to four options, similar to Democracy dilemmas and NationStates issues. Situations are persistent conditions, like unrest, corruption spiral, plague, famine, insurgency, border crisis, or space-race pressure, and they stay present until their stop conditions are met. Chains are authored story arcs that save state across multiple steps and can branch based on earlier outcomes. Democracy’s official docs already distinguish events, dilemmas, and situations this way, with different trigger logic and different persistence models. ¹¹

The cleanest technical pattern is to make every major system emit **standardized world facts**, and let the event manager consume those facts plus the current national state. Policies emit facts like `policy_changed`, `tax_raised`, `civil_freedom_reduced`, or `subsidy_added`. Development emits `district_completed`, `industrialization_up`, or `infrastructure_bottleneck`. Diplomacy emits `treaty_signed`, `embargo_started`, `relation_collapsed`, or `alliance_requested`. Espionage emits `spy_detected`, `intel_leak`, or `covert_success`. War emits `battle_lost`, `casualties_spike`, `occupation_started`, or `front_stabilized`. Later, when you add NPCs, they should use the **same fact interface**. That way you do not need a second event system for AI actors; NPCs simply become additional broadcasters into the same pipeline. Unity’s official event-channel pattern exists for exactly this sort of loose-coupling between systems. ¹²

For selection, I would not copy NationStates or Democracy exactly; I would combine them. NationStates shows the value of a mostly random draw from an eligible pool, which keeps the game surprising, while

Democracy shows the value of scoring every event against the current state and firing the most contextually relevant content. A very effective hybrid is:

```
eligible set
→ remove blocked, obsolete, cooldown, duplicate, and incompatible content
→ score remaining items by context
→ apply novelty, anti-spam, and pacing multipliers
→ sample by weighted randomness inside category quotas
```

That gives you contextual relevance without making the game feel predictable. It also lets you guarantee variety, such as “at most one domestic crisis, one foreign-development headline, and one faction dilemma per turn.” NationStates’ staff describe issue assignment as mostly random among qualifying content, while Democracy 4’s official event manager evaluates every event and triggers the highest-scoring one if it crosses a threshold. ¹³

The scoring model should borrow heavily from Democracy’s official **influences** and **grudges**. A simple but powerful formula is:

```
finalWeight =
    baseWeight
    * triggerContext
    * narrativeNeed
    * novelty
    * antiSpam
    * eraFit
    * severityBudget
    * randomJitter
```

`triggerContext` is the product or average of condition curves: high unemployment may strongly boost labor unrest; low liberty plus high fear may boost secret-police scandals; high technology plus low safety may boost research accidents. `narrativeNeed` is your pacing control: if the game has been peaceful for too long, unrest, scandal, or diplomatic tension can be made more likely. `novelty` favors unseen or long-unseen events, which mirrors NationStates’ bias toward content the player has not recently encountered. `antiSpam` uses cooldown groups and diminishing returns so the same theme does not repeat every other turn. ¹⁴

Consequences should not be applied only as flat instantaneous deltas. Democracy’s official documentation is especially useful here: policies can have **inertia**, meaning they take several turns to fully take effect, and both dilemmas and events can create **decaying grudges**, temporary hidden modifiers that fade over time. Cliff Harris also warned Democracy 2 players that major policy changes can take a year or more to show their full downstream consequences. That is exactly how you should handle event aftermath. A riot should not just mean `Crime +5`; it should also create a decaying fear modifier, a decaying tourism penalty, a slower-moving legitimacy cost, and maybe a long-tail boost to authoritarian sentiment. ¹⁵

This is also where your requested sources of impact fit naturally. **Policy decisions** tend to create inertia and long-tail modifiers. **Player actions** create direct facts and immediate dilemmas. **Development** changes background probabilities. **Diplomacy** changes bilateral and regional event decks. **Economic actions** shift unemployment, trade, inflation, and black-market behavior. **Espionage** creates hidden facts, delayed reveals, and plausible deniability branches. **War** turns many peacetime situations into wartime versions and can spawn aftermath content even after the war ends. The important design choice is to make them all write into the same simulation layer, so that a single underlying change can create different downstream events depending on era, law, culture, or technology.

Era and technology gating

The best way to gate stats and events is with a **single unified prerequisite system** built on hierarchical tags.

At minimum, every piece of content should support `allOf`, `anyOf`, and `noneOf` requirements over tags such as `Era.Medieval`, `Era.Industrial`, `Tech.Rockets`, `Tech.Printing`, `Magic.Enabled`, `Gov.Republic`, `Geo.Coast`, `Geo.LandBorder`, `War.Active`, `Economy.Recession`, or `Religion.StateCult`. This is much cleaner than hard-coding checks inside event scripts. Democracy's official modding docs already use explicit pre-requisites for situations and allow custom prerequisite values to be defined by mods; the developer's forum posts show the same pattern being used to limit events and dilemmas to specific countries or conditions. Unreal's Gameplay Tags system is a very strong implementation reference here because it is hierarchical by design, allowing tags like `Tech.Space.Rockets` to match broader queries like `Tech.Space` without custom code every time. ¹⁶

For your use case, I recommend separating four related but different checks:

- `visible_when`: should the stat, category, or encyclopedia entry appear in the UI at all?
- `eligible_when`: can this content enter the candidate deck?
- `fire_when`: is the current context strong enough that it should actually trigger now?
- `obsolete_when`: should this content be replaced or retired because the world has moved on?

That distinction matters. A **Space Program** stat might become visible the moment Rockets are researched, become actionable only after a launch site exists, and start contributing to actual space-race events only when research capacity and budget meet further thresholds. Likewise, "witch-hunt panic" might become obsolete once rationalist education dominates and magic is disconfirmed, while a broader "moral panic over dangerous new ideas" archetype remains eligible forever but swaps presentation by era.

A very useful refinement is to introduce **reveal states** instead of a binary locked/unlocked switch. I recommend five:

Hidden means the player does not see the stat or event family yet.

Teased means the concept is hinted through rumors, encyclopedia grey boxes, or advisors.

Unlockable means the prerequisite has been met, but the first enabling law, event, or institution has not been established yet.

Visible means the stat now exists in the UI and can be affected.

Actionable means player tools now exist to manipulate it directly.

This is a direct generalization of NationStates’ “Unlockable: Awaiting legislation” and its population-gated custom fields. It also maps cleanly onto your examples: rocketry research can make “Space” unlockable; the first launch facility makes it visible; a formal space agency makes it actionable. ¹⁷

The gating system should also support **replacement and skinning rules** so that one logical archetype can live across many periods. Consider a generic archetype such as `research_test_failure`. In an alchemical era, that might be a lab fire or mana breach. In an industrial era, it might be an explosives-range accident. In a modern era, it becomes a rocket-launch disaster. In a magical setting layered over modernity, it may become an arcane-plasma containment breach. The logic node remains the same—high research intensity plus low safety plus notable secrecy—but the presentation and some sub-effects change with tags.

This matters for stats too. Democracy 4’s official docs explicitly mention situation pre-requisites such as coastline or WMD tech, and official patches show content rules being tuned for specific geography or trigger conditions, such as a border wall policy requiring a land border and the private space program having its trigger behavior adjusted. The lesson is simple: **era, technology, geography, and government form should all be first-class content filters, not hand-authored exceptions buried in prose.** ¹⁸

How to write one event for many eras

The most important authoring rule is to **separate event logic from event presentation**.

Your logic asset should describe scope, prerequisites, scores, outcomes, modifiers, cooldowns, and follow-ups. Your presentation asset should decide the title, body text, option labels, advisor portrait, image, and sound based on tags such as era, culture, religion, tech family, and whether magic exists. That keeps systemic logic reusable and dramatically reduces content bloat.

For text, do not build sentences by stitching together too many fragments. i18next’s localization guidance warns that interpolation causes real translation problems when runtime text concatenates pieces that change grammar, and recommends using multiple self-contained string values whenever possible. Project Fluent’s selector system is a better fit for what you want, because it allows multiple full variants of a message to be selected by an external variable. That means you can switch on `era`, `magicMode`, `cultureGroup`, or `governmentType` cleanly without creating broken or awkward text. ¹⁹

A compact authoring model for one reusable event might look like this:

```
id: research_test_failure
type: dilemma
tags: [Archetype.ResearchAccident, Domain.Science]
visible_when:
  anyOf: [Tech.AdvancedResearch, Magic.Enabled]
eligible_when:
  noneOf: [Era.Ancient]
fire_when:
  allOf: [State.HasResearchSite]
score:
  safety: "0.55 - 0.45*x"
```

```

research_intensity: "0.20 + 0.60*x"
secrecy: "0.05 + 0.20*x"
random: [0.00, 0.15]

presentation:
  titleSelector: era_theme
  titleVariants:
    Alchemy: evt_research_failure_alchemy_title
    Industrial: evt_research_failure_industrial_title
    Rockets: evt_research_failure_rockets_title
    MagicModern: evt_research_failure_arcane_lab_title

  imageSelector: era_theme
  imageVariants:
    Alchemy: art/events/alchemy_lab_fire
    Industrial: art/events/test_range_explosion
    Rockets: art/events/rocket_pad_blast
    MagicModern: art/events/mana_containment_breach

```

That schema is directly inspired by Democracy’s prereqs, influence equations, and temporary-effect scripts, but it uses selector-based presentation to preserve reusability. ²⁰

For images, use the same principle as text: **variant selection with fallback**. If the content has an art asset tagged `Era.Modern + Magic.Enabled`, use that. If not, fall back to `Era.Modern`; if not, fall back to the generic event-art key. Unreal’s Gameplay Tags concept is useful here because it provides a centralized hierarchy for labels and parent-tag matching, which is exactly what you want for asset lookup and fallback chains. ²¹

This structure also lets you write events as **semantic archetypes** instead of era-specific one-offs. “Labor unrest,” “frontier dispute,” “religious schism,” “research accident,” “public panic,” “spy scandal,” and “refugee wave” are all cross-era patterns. What changes is the skin, not the underlying grammar of the event system. That gives you more total content with less maintenance and makes the simulation feel coherent across a long campaign.

Technical implementation blueprint

Use data assets, not hard-coded event logic

The official Positech docs are very clear that almost all simulation elements in Democracy are loaded from simple text or CSV files, and Unity and Godot documentation both make the same broader architectural point: data containers should be separate from scene logic when you want scalable, designer-friendly systems. In Unity, ScriptableObjects are explicitly described as data containers that separate data from logic and can also serve as project-level event channels; in Godot, Resources are described as data containers used by nodes rather than behavior-bearing objects themselves. This is the right architectural direction for your event system whether you use JSON, YAML, ScriptableObjects, Resources, or your own custom asset format. ²²

A practical nation-state event schema would include at least these groups of fields:

```
{
  "id": "frontier_spying_scandal",
  "type": "dilemma",
  "tags": ["Domain.Espionage", "Scope.Foreign", "Tone.Crisis"],
  "visibility": { "allOf": ["Tech.Ciphers"], "noneOf": ["Era.Ancient"] },
  "eligibility": { "allOf": ["State.HasNeighbor"], "noneOf": ["War.Total"] },
  "obsolescence": { "anyOf": ["Tech.PostQuantumCryptography"] },
  "cooldown": { "turns": 24, "group": "espionage_scandals" },
  "scopes": ["nation", "neighbor"],
  "weight": {
    "base": 1.0,
    "influences": [
      { "src": "EspionageExposure", "curve": "0.10 + 0.60*x" },
      { "src": "RelationsWithTarget", "curve": "0.35 - 0.30*x" },
      { "src": "_random_", "min": 0.00, "max": 0.15 }
    ]
  },
  "effects": {
    "onStart": [],
    "onOptionA": [],
    "onOptionB": []
  },
  "followups": ["spy_trial", "quiet_coverup", "border_tension"],
  "presentationKey": "frontier_spying_scandal"
}
```

The point is not the exact field names. The point is that every event should carry **conditions, scoring, persistence, and presentation keys in data** so that new eras, new rulesets, and future NPC systems can plug in without rewriting your core engine.

Run the simulation in a fixed, repeatable pipeline

A solid turn or tick loop looks like this:

```
apply policy inertia
decay temporary modifiers
update hidden substats
recompute visible aggregates
start or stop persistent situations
evaluate unlock and reveal states
ingest world facts from policy / diplomacy / economy / war / espionage
build eligible event deck
score and sample events by category quotas
```



```
present choices and resolve outcomes
schedule follow-ups
record history for graphs, tooltips, and analytics
```

This mirrors both inspirations: NationStates changes eligibility as the nation changes, and Democracy recomputes a large network of values every turn. ²³

Two technical details from Democracy 4 are especially worth copying. First, **compile equations and conditions once at load time** instead of reparsing them every turn. Harris describes a major speedup after caching static parts of evaluation expressions instead of converting them repeatedly at runtime. Second, do a short **simulation warm-up** when starting a new game or loading a scenario with many hidden linked stats; Democracy 4 pre-processes the simulation about 30 times at startup to create a sensible equilibrium and history baseline. If your systems have inertia, lag, and hidden modifiers, a warm-up pass will make the initial world state feel much more believable. ²⁴

Expose trends, reasons, and histories to the player

NationStates' API supports historical census queries, and that is a useful design clue. Players should not only see the current value of Crime or Liberty; they should be able to inspect **why it moved** and **what the trend looks like**. Keep a ring buffer for every important visible stat and for a few hidden diagnostic values. Then let the UI answer questions like:

- "Why did Liberty fall this turn?"
- "What are the top contributors to rising Crime?"
- "Which policy is feeding this situation?"
- "Why is this event eligible now?"
- "Why is this event blocked?"

If NationStates keeps its exact issue-effect formula secret, your game should go in the opposite direction and be more transparent. That will make the complexity feel fairer.

Build authoring and debugging tools early

One of the oldest Positech modding threads ends with Harris saying he hoped to add a toggled event/dilemma debugger, which tells you how important debugging is once a trigger-driven content web becomes large. Do not wait for this. Build it up front. At minimum, every event in your editor should show:

- whether it is hidden, visible, eligible, blocked, or obsolete;
- which prerequisite failed;
- its current raw score and final weighted score;
- which cooldown group is suppressing it;
- what variant text and image keys would be used right now.

A Monte Carlo test mode is also worth the time: run 1,000 simulated campaigns per era or ruleset and log dead content, spam content, average first appearance, average repeat interval, and average impact on each core stat. Harris has written that Democracy's scale is too complex to balance by hand alone, and that he

relies heavily on statistical data and player telemetry to improve late-game balance. Your event system will need the same discipline. ²⁵

Keep the content system ready for future NPCs

Because you plan to add NPC actions later, make the event manager listen to **facts**, not specific actor classes. Today, those facts may come only from the player, the economy system, the war simulator, or scripted world states. Later, an NPC minister, rival nation, megacorp, religion, guild, or intelligence service should be able to publish exactly the same messages into the same system. Unity's event-channel pattern is a good conceptual reference: separate broadcasters from listeners so systems can talk without acquiring brittle direct dependencies. ²⁶

The resulting architecture is straightforward:

- **NationStates-style breadth** for stats, hidden sub-traits, eligibility gates, and state-dependent outcomes.
- **Democracy-style causality** for overlapping systems, delayed consequences, persistent situations, and context-weighted events.
- **Tag-based gating** for eras, technologies, geography, governments, and optional systems like magic or spaceflight.
- **Selector-based presentation** so one event can live across multiple eras without ugly string hacks.
- **Data-driven authoring** so designers can add content without changing engine code.
- **Telemetry and debugging** so the system stays legible and balanceable as it grows. ²⁷

If you implement those pieces together, the game will not just have “more events.” It will have a world model where policy, development, diplomacy, espionage, and war all push on the same underlying machinery, which is what makes a nation sim feel responsive instead of scripted.

¹ ² ¹⁰ ²⁷ NationStates | Frequently Asked Questions

<https://www.nationstates.net/page%3Dfaq>

³ ¹³ ¹⁴ ²³ NationStates • View topic - Are issues selected for you to answer by random selection?

<https://forum.nationstates.net/viewtopic.php?f=15&t=463315>

⁴ Democracy 2 on Steam

https://store.steampowered.com/app/218040/Democracy_2/

⁵ ¹⁶ ²⁰ ²² Democracy 4 - Modding

<https://www.positech.co.uk/democracy4/modding.html>

⁶ ieeee-cog.org

https://ieeee-cog.org/2020/papers/paper_110.pdf

⁷ ⁸ NationStates | The World

<https://dark.nationstates.net/page%3Dworld>

⁹ NationStates • View topic - How are NS issue results calculated?

<https://forum.nationstates.net/viewtopic.php?f=15&t=556705>

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15 Some Democracy 2 hints and tips... – Cliffski's Blog

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